



How hydroponics works

A detailed description of the process of hydroponics, how it works and its applications. <https://digit.in/feb20-07>



Rooftop farming trend in India

Urban and rooftop farming have become a trend in cities such as Bengaluru, Mumbai and Delhi. <https://digit.in/feb20-08>

work, literally. After leaving his University position, Gericke wrote and published 'Complete Guide to Soil-less Gardening' in 1940. Then came into the picture Dennis R. Hoagland, who created a nutrient formula that provided the plants with all the nutrients essential for growth and catered to a large diversity of plant species. The Hoagland solution still serves as the basis for most of the nutrient solutions that are being utilised even today. Alongside Daniel I. Aaron, Hoagland published 'The Water Culture Method for Growing Plants without Soil', which is considered the most important

text in hydroponics culture. There are accounts of the US Army using hydroponics to grow fresh food for the troops stationed in the infertile Pacific Islands, during World War II.

As the hydroponics technology picked up the attention across the world and the usage grew, it's inevitable commercialisation took place. Lawrence Brooks founded 'General Hydroponics' in 1976. Now, even after 40 years, GH remains one of the leaders of continuously expanding the market of hydroponic mineral-nutrient solutions. In 1982, a new area called 'The Land Pavilion' was opened in Disneyworld's EPCOT Center, where a ride goes through "gardens of tomorrow", which are grown using hydroponic technology.

When you think about hydroponics, aside from the typical applications of the practice, you may also think about its applications in space and space travel. NASA is actually quite heavily invested in the practice of hydroponics. A research department completely dedicated to hydroponics and bio-farming was developed by NASA where they researched the growing of plants in extreme environments and hydroponic technology was a big player in the research.

THE PRESENT:

There's no other way to put it, hydroponics is thriving. As per a recent Hydroponics Market Research Report, it is projected that the hydroponics market will be worth \$27.29 billion by the year 2022. Currently,



Dr. William Frederick Gericke, the "father of hydroponics"

almost 90 percent of lettuce and tomatoes that are being grown in the US and Canada are being grown using hydroponics. As the population of the world keeps growing, hydroponics becomes a more viable substitute for traditional farming. Hydroponically grown crops take only 25 percent of the area used by traditional soil growing,

for the same amount of crops. We all know the harrowing threat of water scarcity in the coming years. With hydroponics, water usage can be cut down by 90 percent. This is just another reason why hydroponics is booming. Zero pesticide and herbicide involvement, growth rate twice as fast as traditional agriculture, nutrient solutions with perfect pH eliminating soil worries and the list is never-ending, which is why hydroponics is need for the present.

Countries have already started dedicating acres of lands for commercial hydroponics, including 10,000 acres in Holland, Israel reserving 30,000 acres, Australia and New Zealand having 8,00 acres between them and Canada with hundreds of acres of land for large-scale commercial greenhouses, mainly growing tomatoes, peppers, and cucumber. In India, hydroponics is still a pretty new technology but initiatives are being taken. Many startups have taken it upon them to make hydroponics more popular, affordable and accessible. These startups use new technologies like IoT (Internet of Things), data analysis, etc to perfect the procedure and make it as efficient as possible.

THE UNKNOWN:

While we don't know what the future holds exactly, we can make a couple of calculated predictions as a society. Statistics and reports already indicate that hydroponics will have a special place in the coming years. As the concept of space travel and exploration becomes closer to reality with each passing day, hydroponics will find a place in long flights across space, where agriculture is restricted due to area and soil-weight limits. Similarly, when Mars colonisation does become a reality (Elon, we are looking at you) we won't have to suffer the infertile and barren lands of unexplored planets since hydroponics will allow basic levels of agriculture, which is said to be the foundational step of every human civilization. Several companies across the globe have also begun providing ways to grow vegetables at home as well.

As technological advances are being made, it won't be surprising to see people growing their own assortments of vegetables in their own houses. There are already plenty of homes today that have home hydroponic installations.



Taking 'farming' out of the farm

With the drastic need for more and more land for the ever increasing urban sprawl, which reduces the land available for agriculture, hydroponics will eventually creep into our daily lives and get infused with our culture. Startups, commercial giants and space agencies have already begun traversing into this field. So, don't be surprised, when in the coming years every edible thing you see has hydroponic origins and your neighbour in a high-rise skyscraper is a successful farmer. 🌱